Solutions to Problem Set 3, Second Half

180.301 Microeconomic Theory - Spring 2026

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1. Using backward induction, we first solve the profit maximization problem of Firm B, treating y_A as given.

   $p = 1000 - 0.5y = 1000 - 0.5y_A - 0.5y_B$, Total revenue $TR_B = (1000 - 0.5y_A - 0.5y_B)y_B$. Marginal revenue $MR_B = 1000 - 0.5y_A - y_B$.

   Profit maximization means $MR_B = MC_B$: $1000 - 0.5y_A - y_B = 4y_B \Rightarrow y_B = 200 - 0.1y_A$.

   Now, knowing that Firm B will produce $y_B = 200 - 0.1y_A$, we solve the profit maximization problem of y_A.

   For Firm A, $p = 1000 - 0.5y_A - 0.5y_B = 1000 - 0.5y_A - 0.5(200 - 0.1y_A) = 900 - 0.45y_A$. $MR_A = 900 - 0.9y_A$, $MR_A = MC_A$: $900 - 0.9y_A = 2y_A \Rightarrow y_A^* = \frac{9000}{29}$, $y_B^* = 200 - 0.1y_A^* = \frac{4900}{29}$.

2. In a price-leadership game, Firm A first chooses the price, and then Firm B takes this price as given and chooses its quantity.

   Using backward induction, for Firm B: $p = 1000 - 0.5y_A - 0.5y_B$.

   $0.5y_A = 1000 - p - 0.5y_B \Rightarrow y_A = 2000 - 2p - y_B$. Since Firm B treats price p as given, its profit maximization problem is $p = MC_B$: $p = 4y_B \Rightarrow y_B = \frac{p}{4}$.

   Now, knowing that Firm B will produce $y_B = \frac{p}{4}$, we solve the profit maximization problem of Firm A.

   $p = 1000 - 0.5y_A - 0.5y_B = 1000 - 0.5y_A - \frac{p}{8} \Rightarrow y_A = 2000 - \frac{9p}{4}$. Its profit is $\pi_A = TR_A - TC_A = p(2000 - \frac{9p}{4}) - (2000 - \frac{9p}{4})^2$.

   The derivative with regard to p is $\frac{d\pi_A}{dp} = 2000 - \frac{9p}{2} - 2 \cdot (2000 - \frac{9p}{4}) \cdot (-\frac{9}{4}) = 11000 - \frac{117p}{8} = 0$. Solving, we get $p^* = \frac{88000}{117}$, $y_A^* = 2000 - \frac{9p}{4} = \frac{4000}{13}$, $y_B^* = \frac{p}{4} = \frac{22000}{117}$.

3. Treating Firm A's production y_A as given, Firm B's $MR_B = MC_B$: $y_B = 200 - 0.1y_A$.

   Treating Firm B's production y_B as given, Firm A's $MR_A = 1000 - y_A - 0.5y_B$, $MC_A = 2y_A$, so $MR_A = MC_A \Rightarrow y_A = \frac{1000 - 0.5y_B}{3}$.

   Solving the two equations in the preceding two paragraphs, we get the equilibrium $y_A^* = \frac{18000}{59}$, $y_B^* = \frac{10000}{59}$.

4. It turns out that a Bertrand equilibrium does not exist for this question. The explanation for that is very complex. For the purpose of the exams, you are only required to know the Bertrand equilibrium for firms with constant marginal cost (e.g. $MC_A = 5,\ MC_B = 5$ or $MC_A = 5,\ MC_B = 10$).

5. When the firms form a cartel, they can cooperate and maximize their joint profit.

   First, notice that in a cartel, we must have $MC_A = MC_B$, otherwise production can be shifted from the firm with higher MC to the firm with lower MC to increase profit.

   So, $MC_A = MC_B \ \Rightarrow\ 2y_A = 4y_B \ \Rightarrow\ y_A = 2y_B$.

   Demand curve: $p = 1000 - 0.5y_A - 0.5y_B = 1000 - 1.5y_B$.

   Total profit $\pi = p(y_A + y_B) - y_A^2 - 2y_B^2 = (1000 - 1.5y_B)(2y_B + y_B) - (2y_B)^2 - 2y_B^2 = 3000y_B - 10.5y_B^2$.

   Taking derivative with regard to y_B: $\frac{d\pi}{dy_B} = 3000 - 21y_B = 0 \ \Rightarrow\ y_B^* = \frac{1000}{7},\ y_A^* = \frac{2000}{7},\ p^* = \frac{5500}{7}$.